



Department of Education

STEM Education
National Schools of Excellence
STEM Physics Syllabus
Grades 12
Term 2

<u>DOMAIN</u>	<u>PHYLA</u>
Science	Physics
Technology	
Engineering	
Mathematics	

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Acknowledgements

This STEM Biology Teacher Resource Book for Grade 11 constitutes ten (10) Thematic Areas (Thema) and is intended for students enrolling for STEM Education. In this booklet only three Thema are presented.

Ultima Thulë Ltd was engaged by the Department of Education to develop the STEM Curriculum. The content was generated by writers from universities, secondary schools, new university graduates with knowledge in the subject matter and industry subject experts. Quality assurance groups and the Science Subject Advisory Committee have also contributed to the development of this syllabus.

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SECRETARY'S MESSAGE

This Teacher Guide is to be used exclusively by teachers of Physics to teach Grade 11 Students enrolling for the STEM Education through the Schools of Excellence throughout Papua New Guinea. The STEM Physics Bridging Syllabus ensured that Grade 11 students enrolling for STEM Education were grounded in the basic concepts espoused in Physics before undertaking the advanced concepts introduced in this Teacher Resource Book.

The Teacher Resource Book conforms to the visions of the National Education Plan and the School of Excellence Policy. Furthermore, it is undergirded by a policy and legislative matrix that were issued by the National Government over the last decades, such as the Vision 2050 and Development Strategic Plan 2030. The Preamble of the National Constitution, especially the First Directive of the *National Goals and Directive Principles* makes a clarion call for the Integral Human Development, which the Department took heed of to ensure, through this new syllabus, that *all forms of beneficial creativity, including sciences and cultures, [are] to be actively encouraged*. A principle object of the *National Education Act 1983 (amended 1995)* reiterates this virtue of Integral Human Development (Section 4).

In accordance with the widespread demand for a home-grown curriculum to replace the existing one with the view to position the country as a bastion of innovation in the new century the Secretary of Education has embarked on introducing the STEM Curriculum to the Schools of Excellence. It is hoped that students coming through the STEM Education will not only gain productive employment in the knowledge-based economy but become creators, innovators and inventors of knowledge and ultimately commercialize their creativity through development of their intellectual property rights.

In addition, there are four core elements espoused in this syllabus, insofar as student development is concerned: Students are expected to gain knowledge and skills and apply them to solve every day problems whilst appreciating Physics as a fundamental field of science.

This STEM Physics Teacher Resource Book for Grade 11 constitutes ten (10) Thematic Area (Thema) and is intended for students enrolling for STEM Education. In this booklet only two Thema – Energy and Battery Technology – are presented. By studying these Thema students are expected to gain the foundational knowledge needed in Physics.

The success of this new Curriculum is highly dependent on the Teachers. They play a pivotal role by being innovative and keeping abreast of new information on scientific research and innovations. The challenge for teachers of Physics is to engage students to learn realistic context for increased and better understanding and appreciate the relevance of Physics to human lives, including good health and environment.

In order to ensure Teachers discharge their responsibilities adequately this Teacher Resource Book is accompanied by a Teacher's Guide and Teacher's Resource Book which provides sufficient teaching and assessment materials to help Teachers.

I commend and approve this Teacher Resource Book as the official curriculum for Physics to be used for the Grade 11 STEM Education in the Schools of Excellence throughout Papua New Guinea.

DR UKE KOMBRA, PHD
Secretary for Education

INTRODUCTION

Physics is the study of matter and its composition, how they interact to form new compounds through chemical reactions and the energy they generate as a result of these interactions. The concepts taught in Physics help us to understand these interactions at the molecular and atomic levels. Every substance, whether naturally occurring or artificially produced, consists of one or more elements of the periodic table.

Since Physics is an experimental science, laboratory work becomes an essential part of the syllabus. Practical work can be used to enable students to investigate the properties and reactions of substances and to provide opportunities for students to learn and test the principles and concepts of Physics.

The Grade 11 Physics Teacher Resource Book entails ten (10) Thematic Areas (Thema) and in this Term 4 only three Thema are covered – Earth Science, Infrastructure, and Waste Management. Each Thema has a single Module and several topics. Over time different Modules will be introduced. Since Physics is a highly specialized subject students undertaking Physics are expected to possess high levels of cognitive and numeracy competency and a firm grasp of the language skills.

The modules in this Teacher Guide are sequenced to guide teachers of Physics through the process of imparting basic physics knowledge and skills to students.

RATIONALE

There is a confluence of events that necessitated the need to introduce STEM Curriculum to the Schools of Excellence:

1. The School of Excellence Policy that was launched by Prime Minister Honourable James Marape, MP, in June 2020, calls for the establishment of the Schools of Excellence and the introduction of STEM Curriculum;
2. The Government has decided to replace the Observation Based Education to Standards Based Education in 2019 and tasked the Department of Education to introduce the STEM Curriculum as part of its reform agenda;
3. There is public outcry, both from public and private sectors, over the quality of graduates coming through the National Education System; and
4. The changing landscape in the technology domain globally and the pressing need to produce the next generation of workforce that is agile and can adopt to the changing circumstances as well as become enterprising individuals through their creation of intellectual properties.

The combination of these factors has led the Department of Education to adopt the STEM Curriculum and to introduce this to the Schools of Excellence as a necessary disruptive intervention in an effort to develop the next generation of national intellectual capital and to drive the growth and prosperity of the country through creation of intellectual properties.

In developing the STEM Curriculum, it was realized that the Grades 9 and 10 Science Syllabi does not provide a firm foundation to launch students into STEM Curriculum at the Schools of Excellence. Consequently, a STEM Bridging Program was necessary.

In the Biology Teacher Resource Book students were introduced to the basic concepts and knowledge of Biology in order for them to develop an appreciation of the discipline in a broader context. In this Teacher Resource Book students are introduced to advanced concepts in Biology as they are applied to the ten Thema.

The introduction of the STEM Curriculum, especially the Bridging Program, will cause consequential changes to the Upper Classes of STEM Education and the Lower Secondary Syllabi. This Teacher Resource Book will be introduced in the course of time.

NEW NOMENCLATURE

As part of the overall Curriculum Development new nomenclature were introduced. These are explained below:

Term	Explanation
Domain	Each STEM area is defined as <i>domain</i> (e.g., Science Domain, Technology Domain, etc.)
Phylum	What would normally be regarded as “subject” (e.g., Biology, Physics, Chemistry, etc.) is now regarded as <i>phylum</i> ; the plural is <i>phyla</i> .
Thema	Thematic areas of studies
Module	A specific concept within the Thema which is being studied
Topic	A specific aspect of the Module that is being studied

STEM Domains

At this juncture of our national development there is no other task before us more important than deploying a robust STEM Curriculum that develops the next generation of highly educated students to enter the work force. By building on the best of current practices and standards the STEM Curriculum aims to take us, beyond the constraints of the existing structures of schooling, towards a shared vision of excellence nationally and marketing the nation globally in terms of Science, Technology, Engineering and Mathematics. In this STEM Curriculum, there are ten (10) Thema (Thematic Areas) introduced.

Science

It is the study of the natural world, including the laws of the nature associated with physics, chemistry and biology and the treatment or application of facts, principles, concepts or conventions associated with these disciplines. Science is both a body of knowledge that has been accumulated over time and a process-scientific inquiry-that generates new knowledge. Knowing from science informs the engineering design process. In this STEM Curriculum the Science Domain has three Phyla – Biology, Chemistry and Physics.

Technology

Comprises the entire system of people and organizations, knowledge, processes, and devices that go into creating and operating technological artifacts, as well as the artifacts themselves. Throughout history humans have created technology to satisfy their wants and needs. Much of modern technology is the product of science and engineering and technological tools used in both fields. There are two Phyla – Information Technology and Computer Sciences – introduced in this STEM Curriculum for the Technology Domain.

Engineering

Concerns both the knowledge about the design and creation of products and a process for solving problems. The design process must result in a final product that must operate or function under various constraints. One constraint in engineering design is the laws of nature. Engineers must learn to harness the natural forces at work around them and through their design process develop products that must operate in harmony with nature. Other constraints include things such as time, money, available materials, ergonomics, environmental regulations and manufacturability. Engineering utilizes concepts in science and mathematics as well as technological tools. There are three (3) Phyla introduced in the Engineering Domain – Civil, Mechanical and Electrical.

Mathematics

It is the study of patterns and relationships among quantities, numbers, and shapes. Specific branches of mathematics include arithmetic, geometry, algebra, trigonometry and calculus. Mathematics is used in science and in engineering. In this STEM Curriculum, Mathematics is considered as a Service Domain; specific topics tailored towards specific Thema are also considered relevant to undergird the relevance and applicability of Mathematics Domain to other fields of knowledge.

STEM PHYSICS CONTENT OVERVIEW – GRADE 12, TERM 2

Table 1.1 Sequencing & Benchmarking STEM Physics Content Standards

Setting STEM Physics Content Standards for Progressive Learning & Assessment			
Thema	Energy	Broad Content Standard	
Module 12.1	Nuclear Energy	12.1: Conceptualize physics of nuclear energy and use it to design a simple power plant	
Topics	Performance Indicators	Module Benchmark	Assessment Tasks
1: Introduction to Nuclear Energy	12.1.1a: Clearly define and explain nuclear energy. 12.1.1b: Eloquentlly describe the fundamental particles that constitute to create the atomic structure. 12.1.1c: Define isotopes and describe how the nucleus composition determines whether it is a stable or unstable isotope (radioisotope). 12.1.1d: Define and explain radioactivity and radioactive decay. 12.1.1e: Recognize and distinguish the different types of radioactive decay. 12.1.1f: Explain the law of radioactive decay and half-life and describe why there are different rates of radioactive decay for different radioisotopes. 12.1.1g: Describe Understand the concept of radioactive series.	Students' ability to master the knowledge and skills of; • Atomic structure and fundamental particles • Radiation and Radioactive decay • Nuclear fission and fusion reactions • Nuclear Reactors and general nuclear power plant setup	Research Project + Written Test
2: Nuclear Reactions – Fission and Fusion	12.1.2a: Clearly explain binding energy and how it can determine which substances will undergo nuclear fission and fusion. 12.1.2b: Clearly differentiate nuclear fission and nuclear		

	fusion and list some of the materials used. 12.1.2c: Describe the concept of mass-energy equivalence in a nuclear reaction. 12.1.2d: Calculate the energy released during nuclear fission.		
3: Nuclear Fission Reactors	12.1.3a: Briefly describe the different stages involved in processing uranium for use in fission reactors. 12.1.3b: Define and generally explain how a nuclear power plant works. 12.1.3c: List and explain the different components of a power plant with their operations. 12.1.3d: Differentiate and describe how a boiling water and pressurized water power plant works. 12.1.3e: Describe the efficiency of a nuclear power plant.		
Thema	Battery Technology	Broad Content Standard	
Module 12.1	Inductive Charging	12.1 Grasp the knowledge of electromagnetic induction applied in electronics, particularly Integrating battery technology in Hybrid Electric Vehicles.	
Topics	Performance Indicators	Module Benchmark Students' ability to master the knowledge and skills of;	Assessment Tasks
1: Electromagnetism	12.1.1a: Explain the phenomena of electromagnetism. 12.1.1b: Describe how a magnetic field generates current and vice versa. 12.1.1c: Clearly explain how electromagnets and	- The basic working mechanism of Fossil Powered Vehicles. - The basic working mechanism of Electric Vehicles.	Research Project + Written Test

	transformers utilize electromagnetic induction.	The basic working mechanism of Hybrid Electric Vehicles.	
2: Inductive Charging	12.1.2a: Describe the process involved in Inductive Charging 12.1.2b: How inductive charging is utilized in charging electric vehicles		
3: Hybrid Electric Vehicles	12.1.3a: Differentiate between IC engines and Hybrid Electric Vehicles. 12.1.3b: Identify and distinguish between the different types of Hybrid Electric Vehicles. 12.1.3c: Explicitly explain and describe how hybrid electric vehicles integrate battery technology. 12.1.3d: Differentiate between IC engines and hybrid Electric Vehicles. 12.1.3e: Describe the role of batteries and inductive charging for the future electric vehicle market.		

Table 1.2 Mapping Content Progressions by Modules

Thema/Module	Broad Content Standard	12.1	12.1
Energy/ Module 12.1: Nuclear Energy	12.1 Conceptualize the knowledge physics in the atomic structure that contributes to harnessing nuclear energy 12.2 Grasp the concept of nuclear energy in and its applications in large scale power generation in fission and fusion power plants.	✓	
Battery Technology/ Module 12.1: Inductive Charging	12.1 Grasp the knowledge of electromagnetic induction applied in electronics, particularly Integrating battery technology in Hybrid Electric Vehicles.		✓

THEMA: ENERGY

MODULE 12.1: NUCLEAR ENERGY

4 weeks

Mentioned previously, there are different types of energy – thermal, electrical, solar, hydro, mechanical, chemical, biological, etc. In this module, we will focus on nuclear energy.

Nuclear energy is the energy stored in an atom's nucleus. Every object in the universe is made up of atoms, which are small particles. The bonds that hold these atoms together contain a tremendous amount of energy which can be used to generate electricity. However, the energy must first be released. Nuclear fusion and nuclear fission are two methods for releasing nuclear energy. Nuclear fusion releases energy when atoms unite or fuse together to form a bigger atom and the sun generates energy in this manner. Nuclear fission releases energy when atoms are torn apart to form smaller atoms and it is the most common method for harnessing nuclear energy in which nuclear fission power facilities generate electricity.

In this module, students will learn about nuclear energy, the materials involved in the process of harnessing nuclear energy through fission reaction, the applications of nuclear energy (in terms of nuclear technologies), and specifically look at how nuclear power plants generate electricity.

Prerequisites:

- Grade 11 Term 3, STEM Physics (Thema: ENERGY)
- Grade 11 Term 3, STEM Engineering (Thema: ENERGY)

Bridging Phyla(s):

- NIL

Broad Content Standard

12.1 Conceptualize the knowledge physics in the atomic structure that contributes to harnessing nuclear energy

12.2 Grasp the concept of nuclear energy in and its applications in large scale power generation in fission and fusion power plants.

Content Benchmark

Students' ability to master the knowledge and skills of;

- Atomic structure and fundamental particles
- Radiation and Radioactive decay
- Nuclear fission and fusion reactions
- Nuclear Reactors and general nuclear power plant setup

Topic 1: Introduction to Nuclear Energy

Performance Indicators

Students are expected to;

12.1.1a: Clearly define and explain nuclear energy.

12.1.1b: Eloquently describe the fundamental particles that constitute to create the atomic structure.

12.1.1c: Define isotopes and describe how the nucleus composition determines whether it is a stable or unstable isotope (radioisotope).

12.1.1d: Define and explain radioactivity and radioactive decay.

12.1.1e: Recognize and distinguish the different types of radioactive decay.

12.1.1f: Explain the law of radioactive decay and half-life and describe why there are different rates of radioactive decay for different radioisotopes.

12.1.1g: Describe Understand the concept of radioactive series.

Content Expansion

Body of Knowledge for their understanding

- Introduction
 - Atomic Structure
 - Fermions
 - Bosons
- Isotopes and Nuclear Stability
- Types of Radioactive Decay
 - Alpha Decay
 - Beta Decay
 - Beta-Plus Decay - Positron Emission
 - Beta-Minus Decay
 - Electron Capture
 - Gamma Decay
- Law of Radioactive Decay
- Rate of Radioactive Decay
 - Half-life
- Radioactive Decay Series

Set of Skills for their proficiency levels

- Analytical and computational skills
- Research skills
- Written and oral communication skills
- Observational skills
- Teamwork and interpersonal skills

Trait of Attitudes and Values to govern their lives

- Appreciate the knowledge of the atomic structure and how energy is released through radiation that can be harnessed.
- Develop a logical and independent mind.
- Meticulous attention to detail and accuracy.

Topic 2: Nuclear Reactions - Fission and Fusion

Performance Indicators

Students are expected to;

12.1.2a: Clearly explain binding energy and how it can determine which substances will undergo nuclear fission and fusion.

12.1.2b: Clearly differentiate nuclear fission and nuclear fusion and list some of the materials used.

12.1.2c: Describe the concept of mass-energy equivalence in a nuclear reaction.

12.1.2d: Calculate the energy released during nuclear fission.

Content Expansion

Body of Knowledge for their understanding

- Introduction
- Binding Energy
- Mass-Energy Equivalence
- Nuclear Fission
 - Byproducts of Fission
 - Fissile Materials
- Nuclear Fusion
- Energy Calculation Examples
 - Single Step Radioactive Decay
 - Multi-step Radioactive Decay

Set of Skills for their proficiency levels

- Analytical and computational skills
- Research skills
- Written and oral communication skills
- Observational skills
- Teamwork and interpersonal skills

Trait of Attitudes and Values to govern their lives

- Appreciate the energy released from fission and fusion that are alternate sources for generating electricity.
- Develop a logical and independent mind.
- Meticulous attention to detail and accuracy.

Topic 3: Nuclear Fission Reactors

Performance Indicators

Students are expected to;

12.1.3a: Briefly describe the different stages involved in processing uranium for use in fission reactors.

12.1.3b: Define and generally explain how a nuclear power plant works.

12.1.3c: List and explain the different components of a power plant with their operations.

12.1.3d: Differentiate and describe how a boiling water and pressurized water power plant works.

12.1.3e: Describe the efficiency of a nuclear power plant.

Content Expansion

Body of Knowledge for their understanding

- Introduction to nuclear power plants
- Uranium Mining
- Milling and Processing
- Fission Reactors
 - Light Water Reactors
 - Boiling-Water Reactors
 - Pressurized-Water Reactors
- Nuclear Power Plant
- Energy obtained from Nuclear Reactors

Set of Skills for their proficiency levels

- Ability to plan and design a simple nuclear power plant in the country.
- Analytical and computational skills
- Research skills
- Written and oral communication skills
- Observational skills
- Teamwork and interpersonal skills

Trait of Attitudes and Values to govern their lives

- Appreciate power plants and use the concepts to design a simple power plant for electricity generation.
- Develop a logical and independent mind.
- Meticulous attention to detail and accuracy.

THEMA: BATTERY TECHNOLOGY

Introduction

A battery is a storage device and converter for chemical energy and electrical energy. It has one or more electrochemical cells that are connected to external circuits. The resulting conversion of the stored chemical energy to electrical energy powers the electrical devices such as flashlights, electric cars and mobile phones. When looking at the battery, the positive terminal is known as the cathode while the negative terminal is known anode.

Electrons flow from the anode then to the external circuit and onto the cathode. When the battery is connected in a circuit to give electricity, there is a reaction where electrons are gained and lost and this reaction is known as the reduction-oxidation reaction or in short, the redox reaction. The redox reaction in the battery converts the stored high chemical energy which reacts to low energy products and the free energy difference between the high energy of the reactants and the low energy of the products results as electrical energy to the external circuit.

Listed below are few of the many career pathways for students who are interested in the field of battery technology development.

- Cell Developer
- Process Engineering Technician
- Mechanical Engineer (Battery Performance Testing)
- Battery Product Design Engineer – Cell Mechanical Characterization
- Electrode Coating Process Chemistry/Engineer
- Battery Cell Engineer, and etc.

MODULE 12.1: INDUCTIVE CHARGING

4 weeks

Also known as wireless charging, inductive charging is a method of charging the battery in portable electrical devices without having to plug the device directly into a power socket. In most cases, smartphones were one of the earlier devices that were capable of utilizing this electromagnetic phenomenon.

In recent years, inductive charging is now one of the alternate modes for charging electric vehicles. In this module, we will discuss the concept of electromagnetism and how inductive charging is used to “refuel” electric vehicles.

Prerequisites:

- Grade 11 Term 3, STEM Engineering (Thema: BATTERY TECHNOLOGY)

Bridging Phyla(s):

- NIL

Broad Content Standard

12.1 Grasp the knowledge of electromagnetic induction applied in electronics, particularly Integrating battery technology in Hybrid Electric Vehicles.

Content Benchmark

Students’ ability to master knowledge and skills of;

- Electromagnetism
- Inductive Charging
- Basic mechanism of a Hybrid Electric Vehicle

Content Benchmark

Students’ ability to master the knowledge and skills of;

- The basic working mechanism of Fossil Powered Vehicles.
- The basic working mechanism of Electric Vehicles.
- The basic working mechanism of Hybrid Electric Vehicles.

Topic 1: Electromagnetism

Performance Indicators

Students are expected to;

12.1.1a: Explain the phenomena of electromagnetism.

12.1.1b: Describe how a magnetic field generates current and vice versa.

12.1.1c: Clearly explain how electromagnets and transformers utilize electromagnetic induction.

Content Expansion

Body of Knowledge for their understanding

- Introduction
- Magnetic Field associated with a current
 - Real Life Applications - Electromagnets
- Current Induced by a Changing Magnetic Field
 - Real-life Applications

Set of Skills for their proficiency levels

- Developing a wireless setup of inductive power transfer using a tesla coil and a light bulb.
- Analytical and computational skills
- Research skills
- Written and oral communication skills
- Observational skills
- Teamwork and interpersonal skills

Trait of Attitudes and Values to govern their lives

- Develop a logical and independent mind.
- Meticulous attention to detail and accuracy.

Topic 2: Induction Charging

Performance Indicators

Students are expected to;

12.1.2a: Describe the process involved in Inductive Charging

12.1.2b: How inductive charging is utilized in charging electric vehicles

Content Expansion

Body of Knowledge for their understanding

- Introduction
 - Overview of basic concept of wireless technology
- Working Principle of Inductive Charging
- Inductive Charging in Vehicles
 - Limitations of Inductive Charging
 - Possible Solutions – Electric Roads
 - Major components
 - Copper Coils
 - Receiver Unit
 - Management Unit
 - Cloud Management System

Set of Skills for their proficiency levels

- Research skills
- Written and oral communication skills
- Observational skills
- Teamwork and interpersonal skills

Trait of Attitudes and Values to govern their lives

- Develop a logical and independent mind.
- Meticulous attention to detail and accuracy.

Topic 3: Hybrid Electric Vehicles

Performance Indicators

Students are expected to;

12.1.3a: Differentiate between IC engines and Hybrid Electric Vehicles.

12.1.3b: Identify and distinguish between the different types of Hybrid Electric Vehicles.

12.1.3c: Explicitly explain and describe how hybrid electric vehicles integrate battery technology.

12.1.3d: Differentiate between IC engines and hybrid Electric Vehicles.

12.1.3e: Describe the role of batteries and inductive charging for the future electric vehicle market.

Content Expansion

Body of Knowledge for their understanding

- Fossil Fuel Vs Electric Power
- Fossil Fuel-Electric Structure – Hybrid System
- Types of Hybrid Systems
 - Full Hybrid
 - Mild Hybrid
 - Plug In Hybrid

Set of Skills for their proficiency levels

- Analytical and problem-solving skills
- Research skills
- Written and oral communication skills
- Observational skills
- Teamwork and interpersonal skills

Trait of Attitudes and Values to govern their lives

- Develop a logical and independent mind.
- Meticulous attention to detail and accuracy.

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